Reset motor's DQ

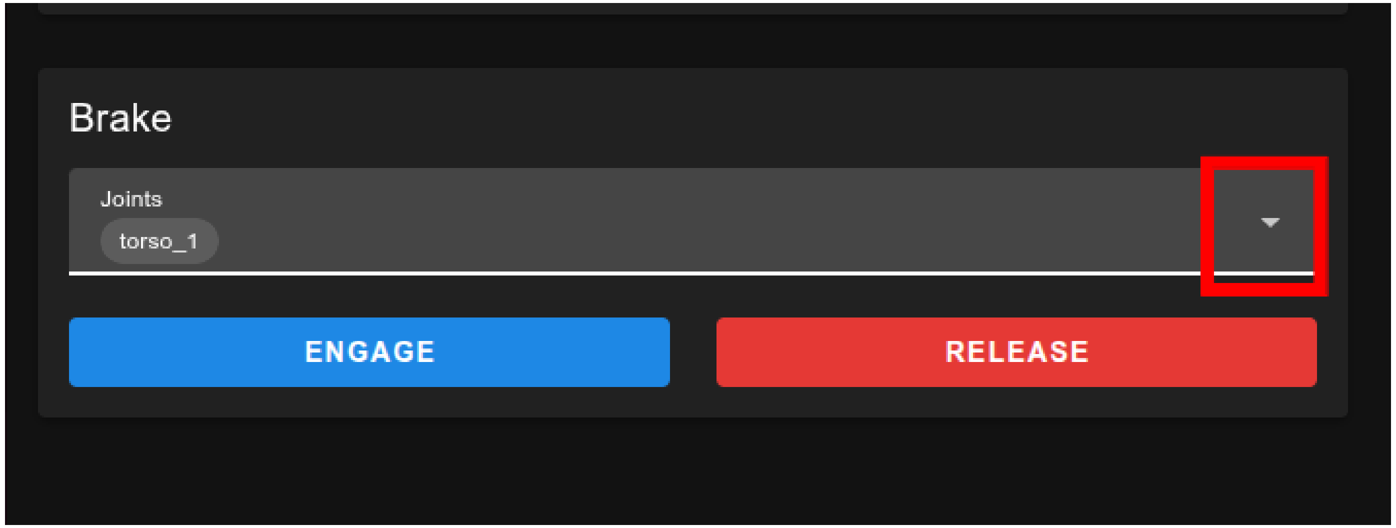
RainbowRobotics

DQ reset manual

setting

To initialize the motor's dq-axis, the robot must meet the following conditions.

- The servo is turned off, and only the 48V power supply is on.
- Release the brake of the motor for which the dq-axis calibration is to be performed
  - Please check the robot's posture by referring to the next slide.



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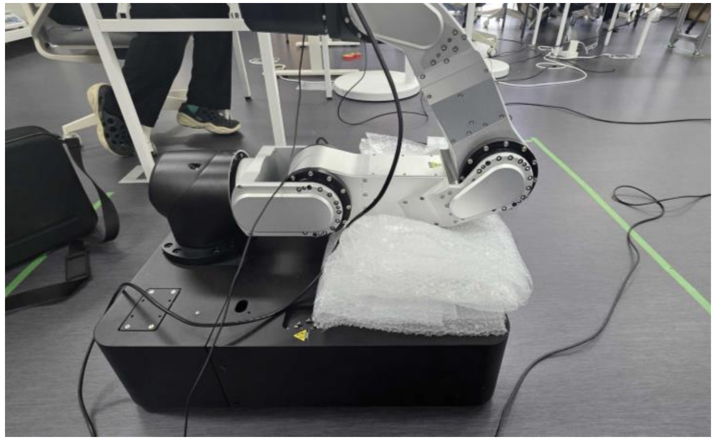
- Ensure that no external load is applied to the motor to be initialized.
  - Upper body
    - Position in the "zero pose" before proceeding
  - Lower body
    - Option 1 : Eliminate the load by using a device capable of suspending the robot, such as a hoist.
    - Option 2 : Please use cushioning material to ensure no load is applied to the motor being initialized.

      Caution : The motor moves slightly during the dq-axis initialization process. Please take care to ensure there are no physical collisions; using a cushioning material is recommended for this reason.

Option 1



Option 2
(torso_1)

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RAINBOW ROBOTICS

File type

- rby1_dq_x86 : use in your pc.
- rby1_dq_jetson : use in upc.

- Code process : reset DQ -> brake engage -> power off -> power on

- Step1 : Download file

- Step2 : authorization
  - chmod +x ./rby1_dq_x86 # or chmod +x ./rby1_dq_jetson

- Step3 : DQ reset(During this process, a backup of the existing dq values is created in the rby1_dq_backup folder).
  - ./rby1_dq_x86 reset <RPC_IP> <Joint Name>

    or
  - ./rby1_dq_jetson reset 192.168.30.1 <Joint Name>
- If the configuration results seem inaccurate, restore the original dq values using the command below.
  - ./rby1_dq_x86 restore <RPC_IP> <Joint Name> --from <BackUp Json File>
  - ./rby1_dq_jetson restore 192.168.30.1 <Joint Name> --from <BackUp Json File>

  ex)./rby1_dq_x86 restore 192.168.1.50 torso_1 –from ./rby1_dq_backup/rby1_dq_torso_1_20260811T030705Z.json

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RAINBOW ROBOTICS

Caution

- There are points during the code execution that require user confirmation. Please keep this in mind as you proceed.

```
are
[WARN] leaves its control mode disabled afterwards. It may not hold again until
the
[WARN] robot is power cycled.
[WARN] When the reset finishes, the tool sets the Control Manager to Idle and
[WARN] power-cycles 48V: the WHOLE robot briefly loses power. (--no-power-cycle
to skip.)
[WARN] No SSH connection and no CAN frame has been sent yet.

Type "yes" to continue [yes]: yes
```

```
#
#   >>> The joint MUST be unloaded and physically supported BEFORE you
#   >>> continue. Keep hands and people clear of its full range of motion.
#
##############################################################################

[WARN] The next step runs DQ align on right_arm_0 and overwrites its
[WARN] stored DQ values. Undoing this needs the backup above.

Type the joint name exactly to proceed [right_arm_0]: right_arm_0
```